PHB 227
Statistical Learning and Multivariate Analysis

UNSUPERVISED LEARNING

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[1]Slides Courtesy Dr. Loki Natarajan

Unsupervised Learning

Goals

In the "unsupervised" setting there are predictors $X_1, X_2, \cdots, X_p$ but no outcome Y. The objective here is to understand relationships among the predictors. Questions of interest may be:

- Is there an informative way to visualize the data?
- Can most of the information be captured using a few linear combinations of the predictors?
- Can I identify subgroups of observations (or predictors) that are similar?

Challenges

- Subjective
- Exploratory
- No clear way to assess "success", since we do not know the true answer

Why do we perform unsupervised learning?

- Understand biological basis of disease (e.g., gene expression in breast cancer)
- Tailoring interventions and identifying barriers
- Advertising/marketing

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Approaches for unsupervised learning

- Principal Components Analysis
  - Dimension Reduction: Find $m < p$ linear combinations of $X_1, X_2, \cdots, X_p$, which capture most of the "information" in the predictors.
  - Linear Surface: Find a lower dimensional linear surface that is closest to the predictors
- Cluster Analysis
  - Identify (latent) homogeneous subgroups in the predictor space.

Population Principal Components

Given a random predictor vector $\mathbf{X}^T = (X_1, X_2, \cdots, X_p)$ find ${\Phi_r}^T = (\phi_{1r}, \phi_{2r}, \cdots, \phi_{pr})$ such that for $r = 1, 2, \cdots, p$

$$Z_r = \phi_{1r}X_1 + \phi_{2r}X_2 \cdots \phi_{pr}X_p = {\Phi_r}^T\mathbf{X}$$

and the following conditions are satisfied:

- $\text{Cov}(Z_j, Z_r) = 0$ for $j < r$
- $\text{var}(Z_r)$ is maximized
- $\sum_{j=1}^{p} \phi_{jr}^2 = 1$

Z_r is referred to as the r-th *principal component and* $\phi_{1r}, \phi_{2r}, \cdots, \phi_{pr}$ *are called* loadings.

How to find Φ's?

Let Σ be the covariance matrix of $\mathbf{X}^T = (X_1, X_2, \cdots, X_p)$. Then the objective is to find Φ_r for $r = 1, 2, \cdots, p$ (with $\sum_{j=1}^{p} \phi_{jr}^2 = 1$) which:

Maximize

$$\text{var}(Z_r) = {\Phi_r}^T \Sigma \Phi_r$$

subject to

$$\text{Cov}(Z_j, Z_r) = \Phi_j^T \Sigma \Phi_r = 0 \text{ for } j \neq r.$$

Theorem: Let $(\lambda_1, \mathbf{e}_1), (\lambda_2, \mathbf{e}_2), \cdots, (\lambda_p, \mathbf{e}_p)$ be eigenvalue-eigenvector pairs for Σ with $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p > 0$, and ${\mathbf{e}_i}^T \mathbf{e}_j = I(i = j)$. Then

1. the r-th principal component is given by $Z_r = \mathbf{e}_r^T \mathbf{X}$
2. $\text{var}(Z_r) = \lambda_r$
3. $\text{Cov}(Z_j, Z_r) = 0$ for $j \neq r$
4. $\sum_{j=1}^{p} \text{var}(X_j) = \sum_{j=1}^{p} \text{var}(Z_j) = \sum_{j=1}^{p} \lambda_j$
5. The proportion of total variance explained by the r-th principal component is $\frac{\lambda_r}{\sum_{j=1}^{p} \lambda_j}$
6. $\text{Corr}(Z_r, X_k) = \frac{e_{rk}\sqrt{\lambda_r}}{\sqrt{\text{var}(X_k)}}$ where e_{rk} is the k-th component of $\mathbf{e}_r$.

Proof: sketch of proof provided in class.

The proof follows from the following two facts from linear algebra.

Fact 1. $\Sigma = P\Lambda P^T = \lambda_1 \mathbf{e_1} \mathbf{e_1}^T + \lambda_2 \mathbf{e_2} \mathbf{e_2}^T + \cdots \lambda_p \mathbf{e_p} \mathbf{e_p}^T$, where P is a matrix whose j-th column is the eigenvector, $\mathbf{e_j}$, $P^T P = I_p$, and Λ is a diagonal matrix with j-th entry equal to the eigenvalue λ_j.

Fact 2. For any positive definite matrix B with eigenvalue-eigenvector pairs $(\lambda_1, \mathbf{e}_1), (\lambda_2, \mathbf{e}_2), \cdots, (\lambda_p, \mathbf{e}_p)$, where $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p > 0$,

$$\max_{\vec{a} \neq 0} \frac{\vec{a}^T B \vec{a}}{\vec{a}^T \vec{a}} = \lambda_1, \text{attained at } \vec{a} = \mathbf{e}_1$$

$$\max_{0 \neq \vec{a} \text{ orthogonal to } \mathbf{e}_1, \mathbf{e}_2, \cdots, \mathbf{e}_k} \frac{\vec{a}^T B \vec{a}}{\vec{a}^T \vec{a}} = \lambda_{k+1}, \text{ attained at } \vec{a} = \mathbf{e}_{k+1}.$$

Interpreting Principal Components Under Multivariate Normality of $\mathbf{X}$

Suppose $\mathbf{X}$ is distributed $\mathcal{N}_p(\mu, \Sigma)$, then we know that the constant density elliposids of $\vec{X}$ are

$$(\mathbf{X} - \mu)^{\mathbf{T}} \mathbf{\Sigma}^{-1} (\mathbf{X} - \mu) = \mathbf{c^2}$$

with axes in the direction of $\mathbf{e_i}$, with lengths $c\sqrt{\lambda_i}$, where the $(\lambda_i, \mathbf{e_i})$ are the eignevalue-eigenvector pairs for Σ. If we assume WLOG that $\mathbf{X}$ is centered (i.e., $\mu = 0$) then

$$\begin{aligned} c^2 &= \mathbf{X}^T \Sigma^{-1} \mathbf{X} \\ &= \frac{1}{\lambda_1} (\mathbf{e_1}^T \mathbf{X})^2 + \frac{1}{\lambda_2} (\mathbf{e_2}^T \mathbf{X})^2 + \cdots + \frac{1}{\lambda_p} (\mathbf{e_p}^T \mathbf{X})^2 \\ &= \frac{1}{\lambda_1} {Z_1}^2 + \frac{1}{\lambda_2} {Z_2}^2 + \cdots + \frac{1}{\lambda_p} {Z_p}^2 \end{aligned}$$

where $Z_1, Z_2, \cdots, Z_p$ are the principal components of $\mathbf{X}$. Thus the PCs lie in the direction of the axes of the constant density elliposoid of $\mathbf{X}$

Principal Components under Multivariate Normality

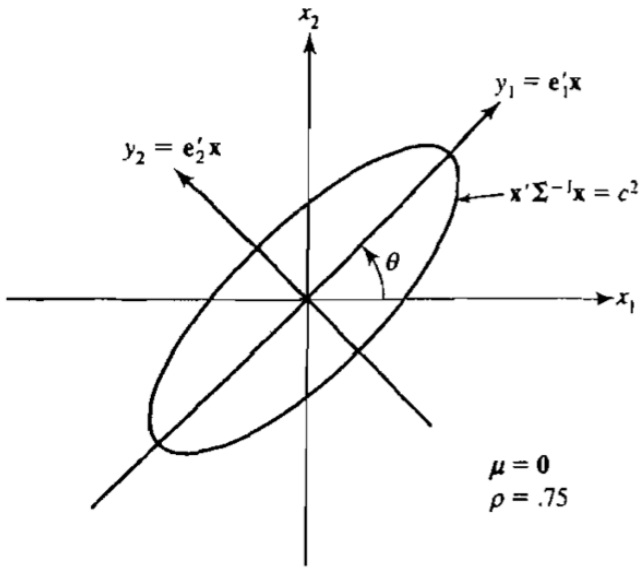


Figure 8.1 The constant density ellipse $\mathbf{x}'\Sigma^{-1}\mathbf{x} = c^2$ and the principal components y_1, y_2 for a bivariate normal random vector $\mathbf{X}$ having mean $\mathbf{0}$.

Reproduced from Johnson & Wichern

Sample Principal Components

Suppose $\vec{x_1}, \vec{x_2}, \cdots \vec{x_n}$, are n independent samples from a p-dimensional distribution with mean μ and variance Σ. Let $\hat{\mu} = \bar{x}$ and $\hat{\Sigma} = S$ be the usual estimates. Let $(\hat{\lambda}_1, \hat{e}_1), (\hat{\lambda}_2, \hat{e}_2), \cdots, (\hat{\lambda}_p, \hat{e}_p)$ be eigenvalue-eigenvector pairs for S with $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \cdots \geq \hat{\lambda}_p$. Then

- the rth sample principal component is given by $\hat{z}_r = \hat{e}_r^T \vec{x}$, where $\vec{x}$ is a random sample drawn from the same distribution, hence has sample mean $\bar{x}$, variance S
- sample variance$(\hat{z}_r) = \hat{\lambda}_r$
- Cov$(\hat{z}_j, \hat{z}_r) = 0$ for $j \neq r$
- Total sample variance$= \Sigma_{j=1}^{p} s_{jj} = \Sigma_{j=1}^{p} \hat{\lambda}_j$
- Sample correlation $r_{\hat{z}_r, x_k} = \frac{\hat{e}_{rk}\sqrt{\hat{\lambda}_r}}{s_{kk}}$.

Example

```
states=row.names(USArrests) #states

apply(USArrests,2,mean)

##   Murder  Assault UrbanPop     Rape
##    7.788  170.760   65.540   21.232

apply(USArrests,2,var)  ##need to scale!

##     Murder    Assault   UrbanPop       Rape
##   18.97047 6945.16571  209.51878   87.72916

pr.out=prcomp(USArrests, scale=TRUE)
names(pr.out)

## [1] "sdev"     "rotation" "center"   "scale"    "x"

cbind(pr.out$center,pr.out$scale,pr.out$scale^2)

##             [,1]      [,2]       [,3]
## Murder     7.788  4.355510   18.97047
## Assault  170.760 83.337661 6945.16571
## UrbanPop  65.540 14.474763  209.51878
## Rape      21.232  9.366385   87.72916
```

Extracting the principal component loadings

```
pr.out$rotation #columns are eigen vectors

##                  PC1        PC2        PC3         PC4
## Murder   -0.5358995  0.4181809 -0.3412327  0.64922780
## Assault  -0.5831836  0.1879856 -0.2681484 -0.74340748
## UrbanPop -0.2781909 -0.8728062 -0.3780158  0.13387773
## Rape     -0.5434321 -0.1673186  0.8177779  0.08902432

pr.out$sdev # sd of PCs, i.e., square-root of eigenvalues

## [1] 1.5748783 0.9948694 0.5971291 0.4164494
```

Each column of the output rotation matrix is the corresponding principal component eigen (loading) vector. In the USArrests example, there are four distinct principal components. In general there are $\min(n-1, p)$

informative principal components in a data set with n observations and p variables.

Extracting the principal component scores

We do not need to explicitly multiply the data by the principal component loading vectors in order to obtain the principal component score vectors. The k-th column of the output matrix x, is the k-th principal component score vector.

```
dim(pr.out$x)

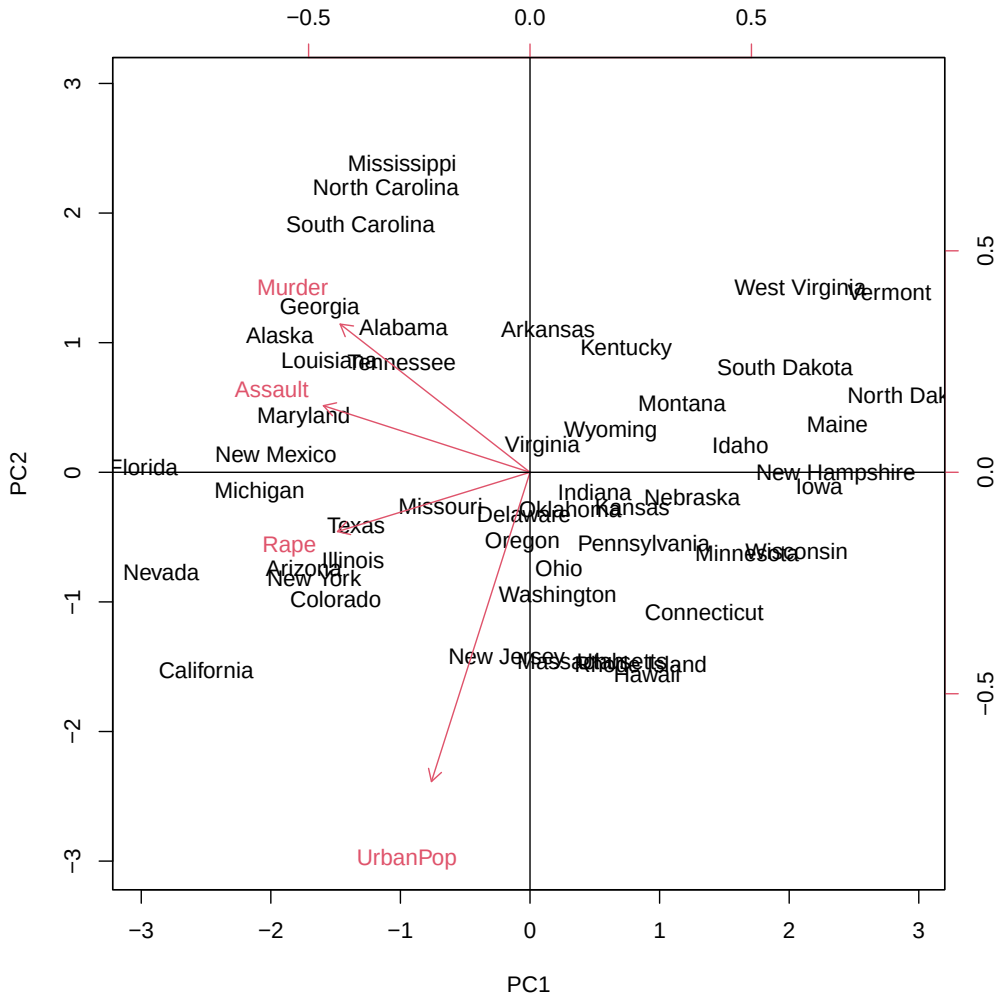
## [1] 50  4

#pr.out£x
```

Recall the PC scores for a sample (i.e., subject or row or state in the USArrests eg) are the projections of that sample onto the eigenvectors.

Visualizing the principal components

```
biplot(pr.out,scale=0)
abline(h=0,v=0)
#biplot(pr.out,scale=0,choices=c(1,3))
```



```
pr.out$center

##   Murder  Assault UrbanPop     Rape
##    7.788  170.760   65.540   21.232

USArrests[c("Virginia","California","Georgia","Vermont","Connecticut"),]

##              Murder Assault UrbanPop Rape
## Virginia        8.5     156       63 20.7
## California      9.0     276       91 40.6
## Georgia        17.4     211       60 25.8
## Vermont         2.2      48       32 11.2
## Connecticut     3.3     110       77 11.1
```

Variance explained by each principal component

```
pr.var=pr.out$sdev^2
pve <- pr.var/sum(pr.var)

pr.var

## [1] 2.4802416 0.9897652 0.3565632 0.1734301

pve

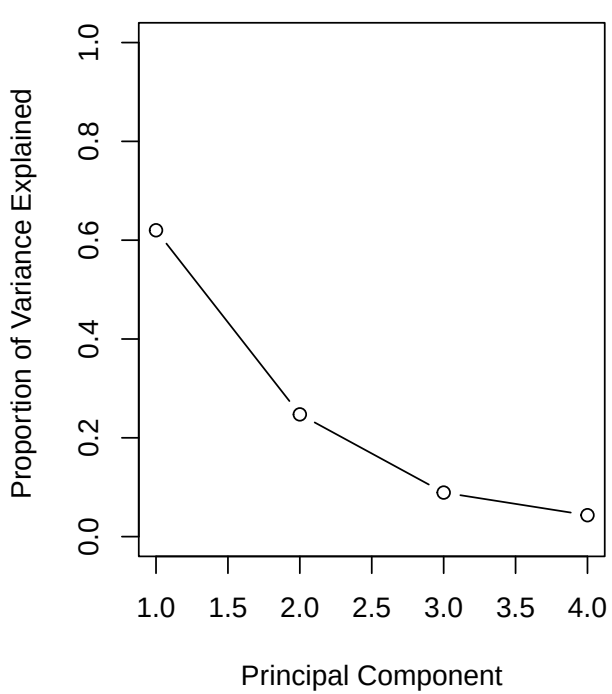
## [1] 0.62006039 0.24744129 0.08914080 0.04335752
```

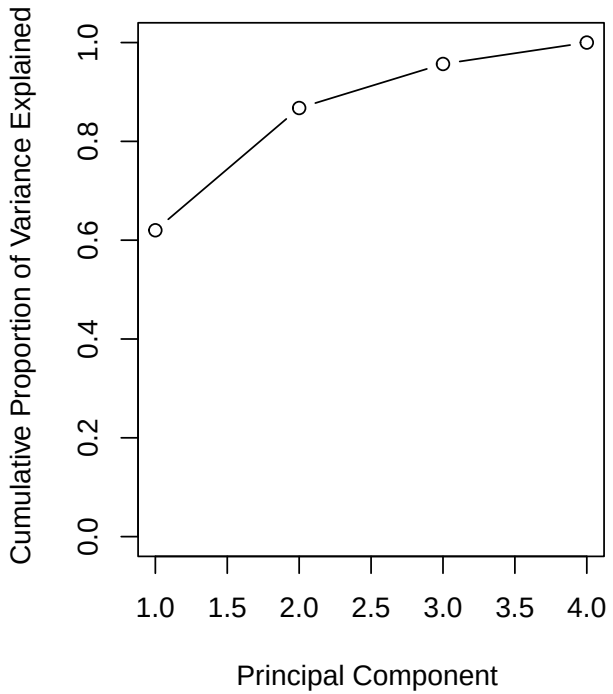
Some considerations in PCA

- How many components?
- Scaling: principal components will change depending on the scale of variables. Hence we usually replace variables by their z-scores, i.e., center and divide by sd, before preforming PCA.
- Uniqueness: principal component loadings and score vector are unique upto a sign flip.

Determining the number of principal components to retain

```
par(mfrow=c(1,2))
plot(pve, xlab="Principal Component", ylab="Proportion of Variance Explained ",
plot(cumsum(pve), xlab="Principal Component ", ylab=" Cumulative Proportion of
```

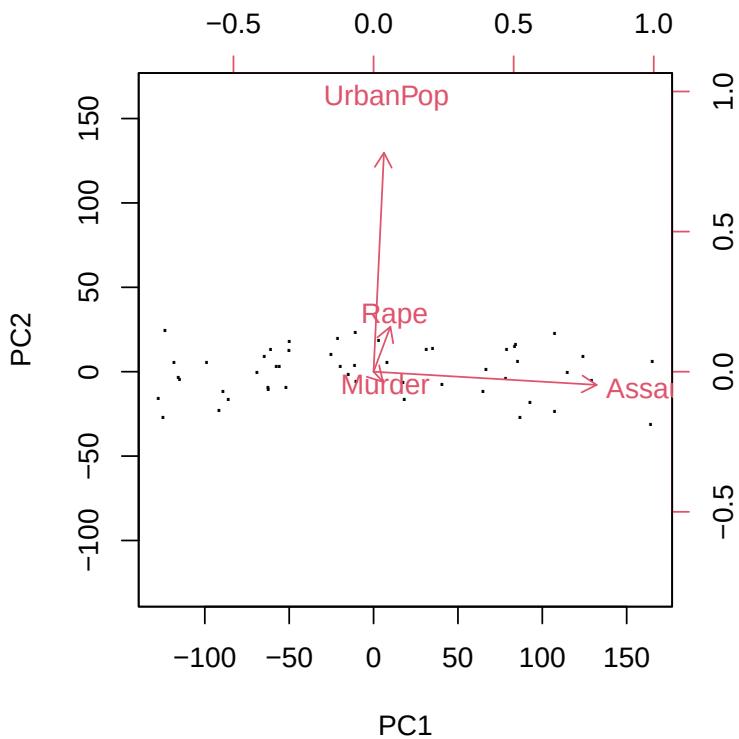




Scaling

```
pr.out2=prcomp(USArrests, scale=FALSE)
```

```
#par(mfrow=c(1,2))
#biplot(pr.out,scale=0)
biplot(pr.out2,scale=0,,xlabs=rep(".", nrow(USArrests)))
```



A Geometric Interpretation

The first principal component (PC) is the line in R^p-space which is *closest* to $\mathbf{X}$. Similarly, the space spanned by the first and second PCs is the plane in R^p-space which is *closest* to $\mathbf{X}$. In general, the vector space spanned by the first r principal components, is the hyperplane closest to $\mathbf{X}$, i.e. $x_{ij} \approx \sum_{m=1}^{r} z_{im}\phi_{jm}$.

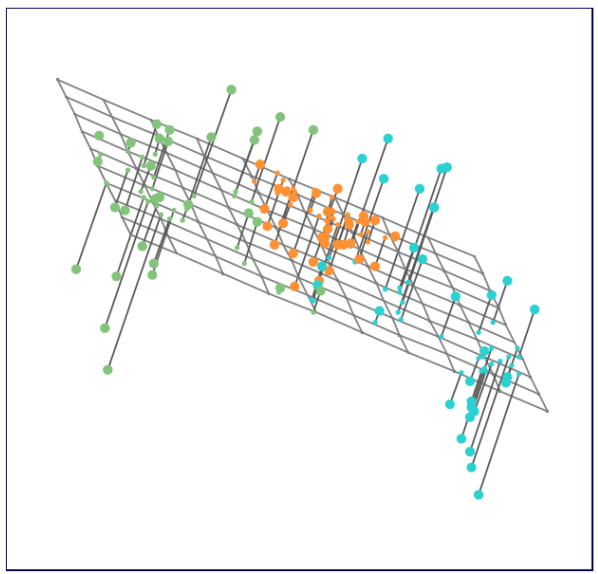


Figure: 90 simulated observations in R^3. The first 2 PCS form the plane which best fits the data.

Geometric Interpretation: Derivation

Notation: If $\vec{u}_1, \vec{u}_2, \cdots, \vec{u}_r$ are r vectors in $\mathcal{R}^p$, then the plane through the origin defined by these vectors $\vec{u}_j$ consists of all linear combinations

$$b_1\vec{u}_1 + b_2\vec{u}_2, \cdots + b_r\vec{u}_r, \text{for some } \vec{b}^T = (b_1, b_2, \cdots, b_r).$$

This plane translated to pass through a point $\vec{c}$ is the plane $\vec{c} + U\vec{b}$, where U is the matrix with columns $\vec{u}_1, \vec{u}_2, \cdots, \vec{u}_r$.

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The objective is to find the best approximating r-dimensional plane to the p-dimensional vectors $\vec{x_1}, \vec{x_2}, \cdots \vec{x_n}$,

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This plane translated to pass through a point $\vec{c}$ is the plane $\vec{c} + U\vec{b}$, where U is the matrix with columns $\vec{u}_1, \vec{u}_2, \cdots, \vec{u}_r$.

The objective is to find the best approximating r-dimensional plane to the p-dimensional vectors $\vec{x_1}, \vec{x_2}, \cdots \vec{x_n}$,

i.e., find the r-dimensional plane $\vec{c} + U\vec{b}$ which minimizes the squared distance between the $\vec{x_j}$s and the plane, i.e., find $\vec{c}$, $\vec{b_j}$, and U which

minimize

$$\Sigma_{j=1}^{n}(\vec{x_j} - \vec{c} - U\vec{b_j})^T(\vec{x_j} - \vec{c} - U\vec{b_j})$$

where $\vec{c}$ is $p \times 1$ $\vec{b_j}$ is $r \times 1$ and U is $p \times r$.

Geometric Interpretation: Derivation

Let $\vec{x_1}, \vec{x_2}, \cdots \vec{x_n}$, be p-dimensional samples with sample covariance S. Let $\hat{e_i}$ be the eigenvectors of S.

Fact: Let $A = [a_1, a_2, \cdots, a_n]^T$ be a $n \times p$ matrix with $\text{rank}(A) = r$. Let $W(A) = \Sigma_1^n (\vec{x_j} - \bar{x} - a_j)^T (\vec{x_j} - \bar{x} - a_j)$. Then, $\hat{A}$ which minimizes $W(A)$ is given by

$$\hat{A} = [(\vec{x_1} - \bar{x}), (\vec{x_2} - \bar{x}), \cdots, (\vec{x_n} - \bar{x})]^T \hat{E}_r \hat{E}_r{}^T$$

where $\hat{E}_r = [\hat{e}_1, \hat{e}_2, \cdots, \hat{e}_r]$. In particular, the j-th column of $\hat{A}^T$ is

$$\hat{a_j} = \hat{y}_{j1}\hat{e}_1 + \hat{y}_{j2}\hat{e}_2 + \cdots + \hat{y}_{jr}\hat{e}_r$$

where $(\hat{y_{j1}}, \hat{y_{j2}}, \cdots, \hat{y_{jr}})^T = [\hat{e_1}^T(\vec{x_j} - \bar{x}), \hat{e_2}^T(\vec{x_j} - \bar{x}), \cdots, \hat{e_r}^T(\vec{x_j} - \bar{x})]^T$, are the coefficients of the first r sample principal components for $\vec{x_j}$.

The geometric interpretation of PCA can be derived from the above fact (see Johnson & Wichern for details).

Geometric Interpretation

In conclusion, the "best" (in least-squares sense) r-dimensional approximation to the p-dimensional $\vec{x}_i$ $(i = 1, 2, \cdots, n)$ is given by the subsapce spanned by the first r eigenvectors $\hat{e}_1, \hat{e}_2, \cdots, \hat{e}_r$ of S. Thus $\vec{x}_i = b_{i1}\hat{e}_1 + b_{i2}\hat{e}_2 + \cdots + b_{ir}\hat{e}_r$ where the coefficients $b_{ij} = \hat{e}_j{}^T(\vec{x}_i - \bar{x})$ are the first r principal component scores of $\vec{x}_i$.

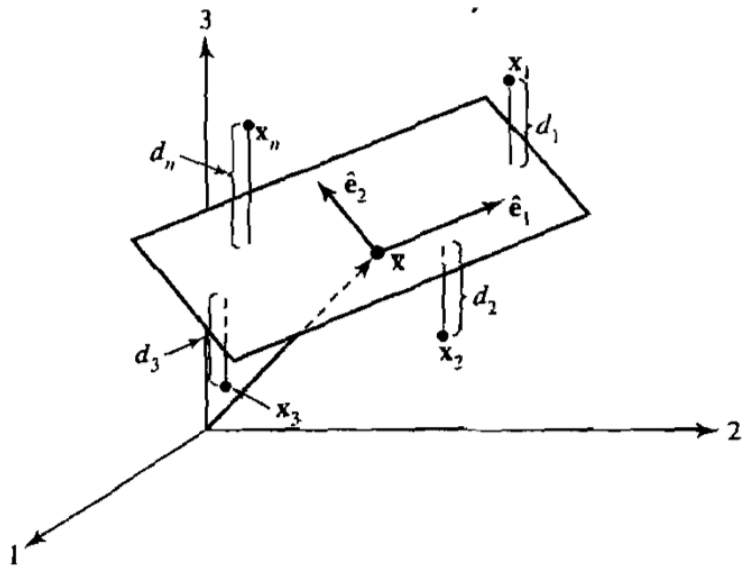


Figure 8.10 The $r = 2$-dimensional plane that approximates the scatter plot by minimizing $\sum_{j=1}^{n} d_j^2$.

Review Questions

1. What is unsupervised learning?
2. What are some applications/examples where unsupervised learning would be useful?
3. What are some pitfalls of unsupervised learning?
4. What is the goal of Principal Component Analysis (PCA)?
5. Describe briefly how we obtain the Principal Components of a set of predictors $X_1, X_2, \cdots, X_p$?
6. What role do eigenvectors and eigenvalues play in PCA?
7. When would a PCA not be successful in reducing dimension?
8. What is a geometric interpretation of PCA?
9. How is PCA diferent from OLS?